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RAJALAKSHMI ENGINEERING COLLEGE

MITRE ATTACK Rule-Based Navigator

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OUTLINE

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ABSTRACT

- Addresses the manual effort of mapping raw system events to the **MITRE ATTACK** framework.
- Presents a rule-based cybersecurity expert system built on a **Goal-Based Intelligent Agent**, **IF-THEN rules, Forward Chaining**, and **Best-First Search**.
- Detects and prioritizes likely attack techniques from observed behaviors such as PowerShell execution, registry modification, or repeated failed logins.
- Outputs the detected technique, risk level, mitigation steps, and reasoning chain.
- Implemented using **classical AI only**.

OBJECTIVE

- To design a **goal-based intelligent agent** that maps observed system behaviors to MITRE ATTACK attack techniques.
- To represent cybersecurity knowledge as **IF-THEN** rules and infer techniques using Forward Chaining.
- To apply **Best-First Search** with the heuristic matched ÷ total conditions to rank rules, so the most strongly supported rule is examined first.
- To output the risk level, mitigation steps, and an **explainable reasoning chain** for each detected technique.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
MITRE ATT&CK: Design and Philosophy	2018	B. E. Strom et al. (MITRE)	Knowledge base mapping observed behaviors to adversary techniques (TTPs)
Detecting Misuse Through Production-Based Expert System (P-BEST)	1999	U. Lindqvist, P. A. Porras	Forward-chaining rule-based expert system for misuse detection
TTPDrill: Extraction of Threat Actions from CTI Sources	2017	G. Husari et al.	Ontology-based mapping of threat actions to ATT&CK techniques
A Formal Basis for the Heuristic Determination of Minimum Cost Paths	1968	P. E. Hart, N. J. Nilsson, B. Raphael	Heuristic best-first search with a priority queue (A*)

EXISTING SYSTEM

- **Signature-based** IDS and antivirus detect only known signatures and **miss behavior-based attacks**.
- Analysts **manually map** suspicious activity to MITRE ATTACK — **slow** and dependent on expert knowledge.
- ML/DL-based detectors act as **black boxes**, offering no interpretable justification for their decisions.
- Alerts often **lack actionable mitigation steps** mapped to a standard framework.
- Traditional rule scanning is **sequential** and does not prioritise the most critical rule first.
- Commercial tools are heavyweight and **hide the underlying reasoning** from the user.

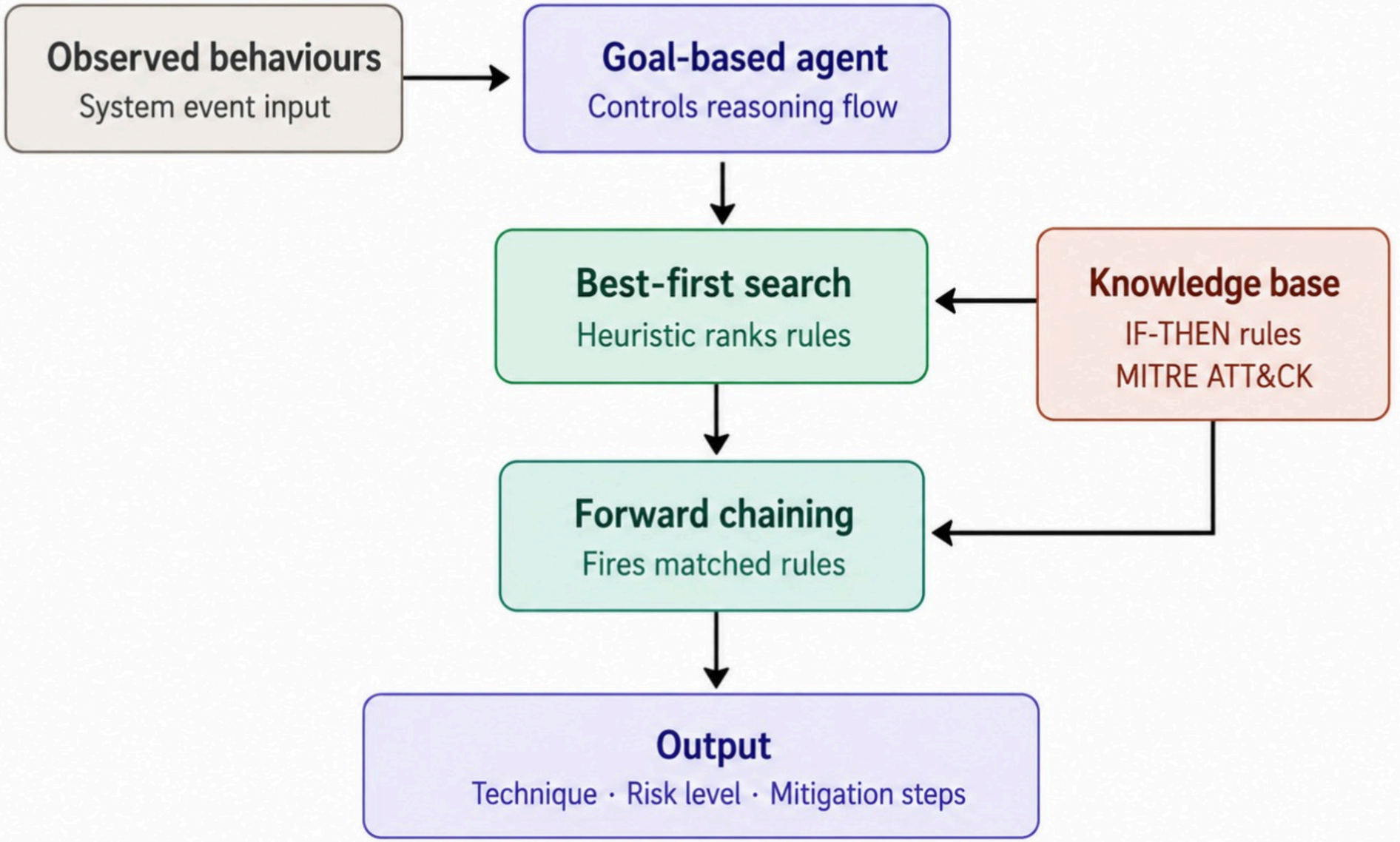
PROPOSED SYSTEM

- A Python rule-based expert system — the **MITRE ATTACK Rule-Based Navigator**.
- A **Goal-Based Intelligent Agent** gathers observed system behaviors and drives reasoning until the goal state — a matched technique — is reached.
- **IF-THEN** rules map behavior sets to techniques: Brute Force, Persistence, Privilege Escalation, Defense Evasion, Data Exfiltration, Initial Access.
- **Best-First Search** with a priority queue ranks rules by a heuristic (matched ÷ total conditions), expanding the most strongly supported rule first.
- **Forward Chaining** fires every rule whose full condition set is satisfied to infer the matching technique(s).
- Outputs the technique, description, **risk level**, **mitigation steps** and **reasoning chain**.

Novelty in Proposed System

- Maps live **observed system behaviors** directly to MITRE ATT&CK techniques with a traceable rule chain, whereas TTPDrill extracts techniques only from written threat reports.
- **Explainable, traceable decisions** through IF-THEN rules, unlike black-box ML models.
- **Heuristic prioritisation** ranks the most strongly supported rule first, instead of a plain sequential scan.
- **Lightweight, Python-only** - no training data, no ML/DL, with mitigation guidance built into every output.

BLOCK DIAGRAM (PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirement:

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum (8 GB recommended)
- Storage: ~2 GB free disk space
- No GPU required

Software Requirement:

- OS: Windows 10 / Linux / macOS
- Python 3.x with Anaconda Navigator / Jupyter Notebook
- Standard libraries only: heapq, csv (no ML/DL libraries)

References

[1] B. E. Strom, A. Applebaum, D. P. Miller, K. C. Nickels, A. G. Pennington, and C. B. Thomas, "MITRE ATT&CK: Design and Philosophy," The MITRE Corporation, 2018 (revised 2020).

[2] U. Lindqvist and P. A. Porras, "Detecting Computer and Network Misuse Through the Production-Based Expert System Toolset (P-BEST)," Proc. IEEE Symposium on Security and Privacy, 1999.

[3] G. Husari, E. Al-Shaer, M. Ahmed, B. Chu, and X. Niu, "TTPDrill: Automatic and Accurate Extraction of Threat Actions from Unstructured Text of CTI Sources," Proc. ACSAC, 2017.

[4] V. Legoy, M. Caselli, C. Seifert, and A. Peter, “Automated Retrieval of ATT&CK Tactics and Techniques for Cyber Threat Reports,” arXiv preprint, 2020.